



Elier Garcia

ELECTRICAL ENGINEER · SITE
SUPERVISION AND ESTIMATING

Contact

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Skills

Electrical Engineering

SAP · Advanced Excel · AutoCAD · OPUS
(Presto equivalent) · LV/MV Installations ·
NEC/IEC/NOM Standards · Site
supervision · Subcontractor management ·
Office 365

Backend & Cloud

Supabase · PostgreSQL · Docker (Self-
hosted)

AI / ML & Data

Gemini AI · RAG Architectures · pgvector ·
Python · SQL

DevOps & Tooling

Power BI · n8n

Languages

Spanish (native)
English B2

Awards & certifications

Electric Power Systems, University at
Buffalo (SUNY) · February 2023

PVC-Coated Conduit Installer, Plasti-
Bond · April 2024

**Artificial Intelligence: Principles &
Applications**, Universidad del Valle de
México · August 2026

Profile

Electrical engineer with 2 years in industrial electrical installations: one estimating and one supervising site work at Microsoft's data center in Querétaro and BMW Group's plant in San Luis Potosí. I specialise in low and medium voltage, materials control and estimating, backed by an M.S. in Business Analytics & AI. I am looking to join a company as an electrical site supervisor or technician, or in an estimating department.

Work experience

Founder · AI & Automation Engineer

2026 — present

EG Solutions · Madrid (remote)

- Design and development of web platforms, n8n automations and AI agents for businesses.

Electrical Site Supervisor

Jan 2024 — Feb 2025

DEMEK S.A. de C.V. · Querétaro / San Luis Potosí, Mexico

Industrial electrical and mechanical contractor in Mexico since 1994: 350+ employees and 900+ contracts.

- **Microsoft Azure hyperscale Data Center** (Querétaro, 9 months): from newest supervisor to **sole lead of the outdoor electrical front** in 6 months, directly supervising **60+ people**, with daily progress meetings, weekly planning and coordination with mechanical, telecom and civil works.
- Outdoor scope: **35 kV medium voltage** with about **13,000 m** of 750 kcmil copper EPR cable in a redundant system (coordinated with civil works), grounding, pump room (MCC, lighting and power panels), outdoor pole lighting and temporary site power.
- Installation, interconnection and quality verification of critical power equipment: **6 diesel generators** (4 × 3,000 kW at 480 V, 1,500 kW and 750 kW), **6 outdoor transformers** (4 × 3,000 kVA, 1,000 kVA and 750 kVA; dry-type and oil-filled), modular UPS, MCCs, switchboards and PDUs, with crane coordination and redundant A/B power cabling.
- **BMW Group plant** (San Luis Potosí, 5 months, 19,847 m² body-shop expansion in 8 sections): supervised LV and grounding in 2 sections (~40 people) and managed **100+ material requisitions** for nearly all 8.
- Built my own materials tracking system (Excel + SAP): every off-budget request technically justified and fewer supply stoppages. It uncovered cable ordered above budget and cable theft, which led to tighter material security.
- Compliance with NEC/IEC/NOM specifications and safety standards on both sites.

Electrical Estimator

Jan 2023 — Jan 2024

DEMEK S.A. de C.V. · Chihuahua, Mexico

- **30+ bids delivered, up to USD 500K**, several in parallel (1-2 week deadlines): drawings and specs in AutoCAD, Excel quantity take-offs by room, system and area, and OPUS pricing (Presto equivalent) by line item.
- Supplier quotes under client-defined brand and function specs, full indirect costs (security, site offices, cranes, labor, safety and documentation) and change orders during execution.
- Won the **Terex project in Nuevo León**: the only bid that reached the final presentation, delivered in English directly to Terex through Copachisa.

Related project

PDF Technical Assistant

pdf-tech-assistant2.vercel.app

Built from my own site experience to query electrical specs and technical manuals: each company uploads its PDFs and gets answers only from the documents, focused on its specialty (electrical, civil, mechanical...). Multi-company with roles, delivered as SaaS or on-premise with Docker.

React 18 · TypeScript · Supabase Edge Functions · pgvector · Gemini

Education

B.S. in Electrical Engineering

Instituto Tecnológico de Chihuahua · Mexico · 2023

Major in electrical installations. GPA 9.27/10. President of the Electrical Engineering student committee.

M.S. in Business Analytics & Artificial Intelligence

INESDI Business TechSchool · Madrid, Spain · 2026

Top academic record of the cohort (8.94/10). ETL/ELT, RAG architectures, applied machine learning and Power BI.